

Conformism–Individualism Trade-offs in LQG Graphon MFG with Control Mean Field Costs[★]

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Abstract: Limitations on the power or resources available to individual agents frequently arise in real-world games. To model such situations, this work studies a class of Linear Quadratic Gaussian Graphon Mean Field Games (LQG–GMFG) whose cost functional incorporates quadratic penalties on deviations from both an agent’s privately desired control and its local control mean field. These penalties represent two distinct motivations: individualism (acting on private preferences) and conformism (avoiding the higher resource costs incurred when acting differently from others). Separate state and control mean-field consistency conditions are imposed, and conditions for the existence of solutions are given. Using spectral decomposition, an explicit value function is obtained for the infinite-horizon, exponentially discounted stationary case, and numerical simulations reveal a trade-off between conformism and individualism.

Keywords: Graphon Mean Field Games, Linear Quadratic Gaussian, Resource Constraints, Value Function Characterization, Networked Multi-Agent Systems

1. INTRODUCTION

In many real-world games, agents often face limitations on the resources or energy they can exploit. Moreover, each agent’s available resources may depend on the actions chosen by all other agents. This interdependence is often treated by imposing shared feasibility constraints, where each agent’s available strategy set is restricted by the collective actions of all other agent; this leads to the notion of Generalized Nash Equilibrium (GNE) (Facchinei and Kanzow, 2010). For dynamic games with a finite set of agents subject to shared constraints the differential variational inequality (DVI) framework has been developed to characterize GNE by Chen and Wang (2014).

In a dynamic game, as the number of agents tends to infinity, the models are called Mean Field Games (MFG), introduced in (Huang et al., 2006). Finding a Nash Equilibrium (NE) in an MFG amounts to identifying a consistent pair comprised of an optimal control and a probability density flow of states. Consistency requires that the optimal control solves the Hamilton–Jacobi–Bellman (HJB) equation parameterized by the density flow, while the flow is reproduced when agents apply the optimal control.

The classical MFG framework has been extended to graphon mean field games (GMFG) to allow for heterogeneous couplings between agents (Caines and Huang, 2021). A graphon can be viewed as the limit of a sequence of finite

graphs, with the limit object defined in the cut metric space induced by the cut norm (Lovász, 2012).

However, the extension of GNE with shared constraints in the MFG or GMFG settings remains largely unexplored. Recently, Hintermüller et al. (2024) considered a GNE problem with mean field interactions under linear deterministic dynamics. At the same time, several works have examined MFGs with state, control, or joint state–control constraints. When the dynamic coupling incorporates the mean field of controls in addition to the mean field of states, the resulting models are termed Mean Field Games of Controls (MFGC) or extended MFGs, with a general analysis presented in (Cardaliaguet and Lehalle, 2018). For MFGC with additional inequality-type mixed constraints, (Bonnans et al., 2023) used a Lagrangian approach to establish existence and uniqueness of an MFGC equilibrium.

The analysis above treats strictly coupled constraints necessitating an equilibrium analysis with interdependent feasible sets. However, an alternative approach is to model these limitations via cost penalization. In this paradigm, the control space remains unconstrained, but agents incur increasingly higher costs as they deviate from a preferred baseline or the collective behavior of the population. Applying this paradigm, the present work introduces control mean field penalties, acting as soft constraints, into the LQG–GMFG framework. By having agents internalize these deviation costs, such a formulation represents the trade-off between individual preference and population-level conformity. In this setting, the main contribution of this paper are:

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- Develop an LQG-GMFG framework where the cost functional incorporates a control mean field penalty. This coupling necessitates separate state and control mean-field consistency conditions, providing the necessary structure to evaluate both state and control empirical mismatches in large finite populations.
- Derive the Forward-Backward ODE (FB-ODE) system that yields the optimal control, making the influence of the control mean-field penalty matrix explicit.
- Obtain an explicit value function formula in the infinite-horizon, exponentially discounted stationary LQG setting.
- Demonstrate the conformism-individualism trade-off via numerical simulations on different graphons.

These contributions provide a foundation for future studies of more challenging problems involving local hard control constraints, nonlinear dynamics and costs, and globally coupled constraints.

2. PRELIMINARIES AND NOTATION

Let $\alpha \in [0, 1]$ be a node index. Assume that the state variable $x_\alpha(t)$ and control $u_\alpha(t)$ are of the same dimension, n . Let $\mathbf{M} : [0, 1]^2 \rightarrow [-c, c]$ denote a graphon, i.e., a bounded symmetric Lebesgue measurable function. The space of all such functions is denoted by $\mathcal{W}_c$. Assume that at each time instant t , the state variables $(x_\alpha(t))_{\alpha \in [0, 1]}$ and the control $(u_\alpha(t))_{\alpha \in [0, 1]}$ are in the Hilbert space $(L^2[0, 1])^n$ with the norm $\|\mathbf{v}\|_{(L^2[0, 1])^n} = \left(\int_{[0, 1]} \|\mathbf{v}(\alpha)\|_{\mathbb{R}^n}^2 d\alpha \right)^{\frac{1}{2}}$. We use $L^2([0, T]; (L^2[0, 1])^n)$ to denote a Hilbert space of equivalence classes of strongly measurable (in the B ocher sense) mappings from $[0, T]$ to $(L^2[0, 1])^n$ that are integrable with the norm $\|\mathbf{x}\|_{L^2([0, T]; (L^2[0, 1])^n)} = \left(\int_0^T \|\mathbf{x}(t)\|_{(L^2[0, 1])^n}^2 dt \right)^{\frac{1}{2}}$.

We can associate a linear bounded operator $\mathbb{T} \in \mathcal{L}((L^2[0, 1])^n)$ with each graphon. For a matrix $D \in \mathbb{R}^{n \times n}$ and $\mathbf{M} \in \mathcal{W}_c$ the operator $[D\mathbf{M}]$ is then defined as follows,

$$([D\mathbf{M}]\mathbf{v})(\alpha) \triangleq D \begin{pmatrix} (\mathbf{M}v_1)(\alpha) \\ \vdots \\ (\mathbf{M}v_n)(\alpha) \end{pmatrix} \in \mathbb{R}^n. \quad (1)$$

Consider the time-varying graphon dynamical system,

$$\dot{\mathbf{x}}(t) = [A(t)\mathbb{I} + D(t)\mathbf{M}]\mathbf{x}(t) + [B(t)\mathbb{I} + E(t)\mathbf{M}]\mathbf{u}(t) \quad (2)$$

where the matrix functions are assumed to be continuous. Let $\mathbb{A}(t) = A(t)\mathbb{I} + D(t)\mathbf{M}$ and $\mathbb{B}(t) = B(t)\mathbb{I} + E(t)\mathbf{M}$. A mild solution to (2) is continuous over $[0, T]$ and satisfies

$$\mathbf{x}(t) = \mathbf{x}_0 + \int_0^t \mathbb{A}(\tau)\mathbf{x}(\tau) + \mathbb{B}(\tau)\mathbf{u}(\tau) d\tau. \quad (3)$$

Consider the initial value problem,

$$\dot{\mathbf{x}}(t) = \mathbb{A}(t)\mathbf{x}(t), \quad \mathbf{x}(s) = \mathbf{x}_s \quad (4)$$

the classical solution to the problem exists by the assumption.

Definition 1. (Evolution operator) The evolution operator associated with $\mathbb{A}$ is defined as the two-parameter family of operators $\Phi(\cdot, \cdot)$ satisfying $\Phi(t, s)\mathbf{x}_s = \mathbf{x}_t$, where $\mathbf{x}(t)$ is the classical solution to (4).

By (Gao et al., 2023, Lem.1), the unique mild solution to (2) is given by

$$\mathbf{x}(t) = \Phi(t, 0)\mathbf{x}_0 + \int_0^t \Phi(t, \tau)[B(\tau)\mathbb{I} + E(\tau)\mathbf{M}]\mathbf{u}(\tau) d\tau. \quad (5)$$

If in addition $\mathbf{u}$ is continuous in $[0, T]$, the solution given by (5) is also a classical solution by (Gao et al., 2023).

3. LQG-GMFG EQUATIONS WITH CONTROL MEAN FIELD COST

This section addresses an LQG-GMFG problem with a control mean-field cost. In this formulation, time-varying penalty matrices weigh a private control's deviations from both its desired signal and the local control mean field.

3.1 A LQG-GMFG Problem with Control Mean Field Cost

Let the dynamics of a generic agent located at node α be:

$$dx_\alpha(t) = (Ax_\alpha(t) + Bu_\alpha(t) + D_0\bar{x}_\alpha(t) + Dz_\alpha(t))dt + \Sigma dw_\alpha(t), \quad x_\alpha(0) = x_{\alpha,0} \sim \mathcal{N}(m_\alpha, \nu^2), \quad (6)$$

where $\bar{x}_\alpha(t)$ denotes the local average of the state variable at node α at time t (adopting the definition established in (Gao and Caines, 2021); $z_\alpha(t) := \int_0^1 \mathbf{M}(\alpha, \beta)\bar{x}_\beta(t) d\beta$ denotes the local state network mean field, and $\mathbf{M}$ is the underlying graphon. For subsequent use in the cost function, the local control mean field $\bar{u}_\alpha(t)$ is defined as the local average of the control variable, analogous to $\bar{x}_\alpha(t)$.

Let the cost function of agents at node α be given by,

$$\begin{aligned} J_\alpha(x_{\alpha,0}, u_\alpha(\cdot); \bar{x}_\alpha(\cdot), z_\alpha(\cdot), \bar{u}_\alpha(\cdot)) \\ = \mathbb{E} \left[\int_0^T \|x_\alpha(t) - \nu_\alpha(t)\|_Q^2 + \|u_\alpha(t) - \xi_\alpha(t)\|_{R_1(t)}^2 \right. \\ \left. + \|u_\alpha(t) - \bar{u}_\alpha(t)\|_{R_2(t)}^2 dt \right. \\ \left. + \|x_\alpha(T) - \nu_\alpha(T)\|_{Q_T}^2 \mid x_\alpha(0) = x_{\alpha,0} \right] \quad (7) \end{aligned}$$

where $\nu_\alpha(t) := H_0\bar{x}_\alpha(t) + Hz_\alpha(t) + \eta$ and $\xi_\alpha(t)$ are the reference state and idiosyncratic control targets, respectively, while $\bar{u}_\alpha(t)$ represents the collective action level of the local population. The matrices Q, Q_T are positive semi-definite, whereas $R_1(t)$ and $R_2(t)$ are uniformly positive definite for all $t \in [0, T]$. Conceptually, $R_1(t)$ and $R_2(t)$ balance individualism and conformism by penalizing deviations from the private target and collective behavior, respectively.

By the Dynamic Programming Principle, the value function $V^\alpha(t, x_\alpha; \bar{x}_\alpha(\cdot), z_\alpha(\cdot), \bar{u}_\alpha(\cdot))$ solves the following HJB equation (Yong and Zhou, 1999, Chap.4, Prop.3.5):

$$\begin{aligned} & -\partial V_t^\alpha(t, x_\alpha; \bar{x}_\alpha(\cdot), z_\alpha(\cdot), \bar{u}_\alpha(\cdot)) \\ & = \inf_{u_\alpha \in U} \left\{ \|x_\alpha - \nu_\alpha(t)\|_Q^2 + \|u_\alpha - \xi_\alpha(t)\|_{R_1(t)}^2 \right. \\ & \quad \left. + \|u_\alpha - \bar{u}_\alpha(t)\|_{R_2(t)}^2 \right. \\ & \quad \left. + (Ax_\alpha + Bu_\alpha + D_0\bar{x}_\alpha(t) + Dz_\alpha(t))^T \right. \\ & \quad \left. \partial V_{x_\alpha}^\alpha(t, x_\alpha; \bar{x}_\alpha(\cdot), z_\alpha(\cdot), \bar{u}_\alpha(\cdot)) \right\} \\ & \quad + \text{tr} \left\{ \Sigma \Sigma^T \partial^2 V_{x_\alpha x_\alpha}^\alpha(t, x_\alpha; \bar{x}_\alpha(\cdot), z_\alpha(\cdot), \bar{u}_\alpha(\cdot)) \right\}. \quad (8) \end{aligned}$$

We use the Ansatz,

$$\begin{aligned} V^\alpha(t, x_\alpha; \bar{x}_\alpha(\cdot), z_\alpha(\cdot), \bar{u}_\alpha(\cdot)) &= x_\alpha^T \Pi_\alpha(t) x_\alpha \\ &+ 2s_\alpha(t)^T x_\alpha + r_\alpha(t). \quad (9) \end{aligned}$$

By the Stochastic Verification Theorem, (Yong and Zhou, 1999, Chap.5, Thm4.1), a control $u_\alpha(t), t \in [0, T]$ is optimal if and only if it minimizes the Hamiltonian. Thus, the optimal control is given by,

$$u_\alpha^o(t) = [R_1(t) + R_2(t)]^{-1} \{R_1(t)\xi_\alpha(t) + R_2(t)\bar{u}_\alpha(t) - B^T[\Pi_\alpha(t)x_\alpha(t) + s_\alpha(t)]\}. \quad (10)$$

Control Mean Field Consistency Condition

Imposing the Consistency Condition by taking expectations on both sides of (10) and combining terms with $\bar{u}_\alpha(t)$, we obtain,

$$\bar{u}_\alpha(t) = \xi_\alpha(t) - R_1(t)^{-1}B^T[\Pi_\alpha(t)\bar{x}_\alpha(t) + s_\alpha(t)]. \quad (11)$$

Substituting (11) back into (10), we can eliminate the term with $\bar{u}_\alpha(t)$ in the optimal control expression to obtain

$$\begin{aligned} u_\alpha^o(t) = & \xi_\alpha(t) - [R_1(t) + R_2(t)]^{-1}B^T\Pi_\alpha(t)x_\alpha(t) \\ & - [R_1(t) + R_2(t)]^{-1}R_2(t)R_1(t)^{-1}B^T\Pi_\alpha(t)\bar{x}_\alpha(t) \\ & - R_1(t)^{-1}B^Ts_\alpha(t). \end{aligned} \quad (12)$$

Substituting (12) into the HJB equation (8), expanding all terms, and equating coefficients of terms with the same order of $x_\alpha(t)$, we get ODEs for $\Pi_\alpha(t)$ and $s_\alpha(t)$,

$$\begin{aligned} -\dot{\Pi}_\alpha(t) = & Q + \Pi_\alpha(t)A + A^T\Pi_\alpha(t) \\ & - \Pi_\alpha(t)B[R_1(t) + R_2(t)]^{-1}B^T\Pi_\alpha(t), \\ \Pi_\alpha(T) = & Q_T. \end{aligned} \quad (13)$$

$$\begin{aligned} \dot{s}_\alpha(t) = & -[A - BR_1(t)B^T\Pi_\alpha(t)]^Ts_\alpha(t) \\ & + \{\Pi_\alpha(t)B[R_1(t) + R_2(t)]^{-1}R_2(t)R_1(t)^{-1}B^T\Pi_\alpha(t) \\ & + QH_0 - \Pi_\alpha(t)D\}\bar{x}_\alpha(t) - \Pi_\alpha(t)B\xi_\alpha(t) \\ & + [QH - \Pi_\alpha(t)D]z_\alpha(t) + Q\eta, \\ s_\alpha(T) = & -Q_T\nu_\alpha(T). \end{aligned} \quad (14)$$

Since ODEs of $\Pi_\alpha(t)$ are the same for all $\alpha \in [0, 1]$, we set $\Pi(t) := \Pi_\alpha(t), \alpha \in [0, 1]$.

State Mean Field Consistency Condition

By taking expectation on both sides of the dynamics of the state (6), and substituting (11) into it, we obtain the ODE for the state mean field,

$$\begin{aligned} \dot{\bar{x}}_\alpha(t) = & [A - BR_1(t)B^T\Pi(t) + D_0]\bar{x}_\alpha(t) \\ & - BR_1(t)^{-1}B^Ts_\alpha(t) + B\xi_\alpha(t) + Dz_\alpha(t), \\ \bar{x}_\alpha(0) = & m_\alpha. \end{aligned} \quad (15)$$

3.2 A Sufficient Condition for the Existence of the Global FB-ODE System

Assume that a graphon $\mathbf{M}(\alpha, \beta)$ is in the space $L^2([0, 1]^2)$; then its associated operator is compact. (Conway, 2019, Chap.2, Prop.4.7) Moreover, since a graphon operator is self-adjoint, it can be expressed via a countably infinite sum of orthogonal projection operators onto its corresponding eigenspaces, each paired with its respective eigenvalue (Conway, 2019, Chap.2, Thm.5.1).

Thus, the global forward-backward equations are,

$$\begin{aligned} \dot{\bar{\mathbf{x}}}(t) = & [(A - BR_1(t)B^T\Pi(t) + D_0)\mathbb{I}]\bar{\mathbf{x}}(t) \\ & - [BR_1(t)^{-1}B^T\Pi]\mathbf{s}(t) + [B\Pi]\xi(t) + [D\mathbf{M}]\bar{\mathbf{x}}(t), \\ \bar{\mathbf{x}}(0) = & \mathbf{m}. \end{aligned} \quad (16)$$

$$\begin{aligned} \dot{\mathbf{s}}(t) = & -[(A - BR_1(t)B^T\Pi(t))^T\mathbb{I}]\mathbf{s}(t) \\ & + [(\Pi(t)B(R_1(t) + R_2(t))^{-1}R_2(t)R_1(t)^{-1}B^T\Pi(t) \\ & + QH_0 - \Pi(t)D)\mathbb{I}]\bar{\mathbf{x}}(t) - [\Pi(t)B\Pi]\xi(t) \\ & + [(QH - \Pi(t)D)\mathbf{M}]\bar{\mathbf{x}}(t) + [Q\Pi](\eta\mathbf{1}), \\ \mathbf{s}(T) = & -[Q_TH_0\mathbb{I} + Q_T\mathbf{H}\mathbf{M}]\bar{\mathbf{x}}(T) - [Q_T\Pi](\eta\mathbf{1}). \end{aligned} \quad (17)$$

We now present a Contraction Principle argument in the space $C([0, T]; (L^2[0, 1]^2))$ which gives a unique solution to (16) and (17).

Let the evolution operators of $[(A - BR_1(t)^{-1}B^T\Pi(t) + D_0)\mathbb{I} + D\mathbf{M}\mathbb{I}]$ and $[-(A - BR_1(t)^{-1}B^T\Pi(t))^T\mathbb{I}]$ be $\phi_1^{\mathbf{M}}(\cdot, \cdot)$, and $\phi_2(\cdot, \cdot)$ respectively. Let $\|\cdot\|_{op}$ denote the operator norm for $\mathcal{L}((L^2[0, 1])^n)$. Define the function as,

$$\begin{aligned} L_0(\mathbf{M}) = & \sup_{t \in [0, T]} \left\{ \int_0^t \|\phi_1^{\mathbf{M}}(t, \tau)[BR_1(\tau)^{-1}B^T\Pi]\phi_2(\tau, T) \right. \\ & [Q_TH_0\mathbb{I} + Q_T\mathbf{H}\mathbf{M}]\|_{op} d\tau \Big\} + \sup_{t \in [0, T]} \left\{ \int_0^t \int_r^T \|\phi_1^{\mathbf{M}}(t, \tau) \right. \\ & [BR_1(\tau)^{-1}B^T\Pi]\phi_2(\tau, r) \Big[(\Pi(r)B(R_1(r) + R_2(r))^{-1} \\ & R_2(r)R_1(r)^{-1}B^T\Pi(r) + QH_0 - \Pi(r)D_0)\mathbb{I} \\ & \left. \left. + (QH - \Pi(r)D)\mathbf{M} \right] \right\|_{op} dr d\tau \Big\}. \end{aligned} \quad (18)$$

Proposition 2. The LQG-GMFG global equations (17) and (16) has a unique classical solution if $L_0(\mathbf{M}) < 1$.

Proof. The classical solutions to the FB-ODE systems are given by the following convolution formula,

$$\begin{aligned} \bar{\mathbf{x}}(t) = & \phi_1^{\mathbf{M}}(t, 0)\mathbf{m} + \int_0^t \phi_1^{\mathbf{M}}(t, \tau) \Big([(-BR_1(\tau)^{-1}B^T)\mathbb{I}] \\ & \mathbf{s}(\tau) + [B\Pi]\xi(\tau) \Big) d\tau. \end{aligned} \quad (19)$$

$$\begin{aligned} \mathbf{s}(\tau) = & \phi_2(\tau, T)\mathbf{s}(T) \\ & - \int_r^T \phi_2(\tau, r) \Big([(\Pi(r)B(R_1(r) + R_2(r))^{-1}R_2(r)R_1(r)^{-1} \\ & B^T\Pi(r) + QH_0 - \Pi(r)D)\mathbb{I} + (QH - \Pi(r)D)\mathbf{M}]\bar{\mathbf{x}}(r) \\ & - [\Pi(r)B\Pi]\xi(r) + [Q\Pi](\eta\mathbf{1}) \Big) dr. \end{aligned} \quad (20)$$

Substituting $\mathbf{s}(\tau)$ in (20) to (19) yields,

$$\begin{aligned} \bar{\mathbf{x}}(t) = & \phi_1^{\mathbf{M}}(t, 0)\mathbf{m} - \int_0^t \phi_1^{\mathbf{M}}(t, \tau) [(BR_1(\tau)^{-1}B^T)\mathbb{I}] \\ & \Big(\phi_2(\tau, T) (-[Q_TH_0\mathbb{I} + Q_T\mathbf{H}\mathbf{M}]\bar{\mathbf{x}}(T) - [Q_T\Pi](\eta\mathbf{1})) \\ & - \int_r^T \phi_2(\tau, r) \Big([(\Pi(r)B(R_1(r) + R_2(r))^{-1}R_2(r)R_1(r)^{-1} \\ & B^T\Pi(r) + QH_0 - \Pi(r)D)\mathbb{I} + (QH - \Pi(r)D)\mathbf{M}]\bar{\mathbf{x}}(r) \\ & - [\Pi(r)B\Pi]\xi(r) + [Q\Pi](\eta\mathbf{1}) \Big) dr \Big) d\tau \\ & + \int_0^t \phi_1^{\mathbf{M}}(t, \tau) [B\Pi]\xi(\tau) d\tau. \end{aligned} \quad (21)$$

Neglecting all terms in (21) independent of $\bar{\mathbf{x}}(t)$, a standard contraction argument yields the result. $\square$

3.3 Spectral Decomposition of the FB-ODE System

Assumption 3. The eigenfunctions $\{\mathbf{f}_\ell\}_{\ell \in \mathcal{I}_\lambda}$ form a complete orthonormal basis in $L^2[0, 1]$, i.e., it is complete.

Remark 4. This completeness assumption ensures that the eigenfunctions span $L^2[0, 1]$ (no nontrivial complement). The UAG graph meets this requirement.

Remark 5. Since $L^2[0, 1]$ is a separable Hilbert space, we can index $\{\mathbf{f}_\ell\}_{\ell \in \mathcal{I}_\lambda}$ by ℓ from 1 to ∞ .

Next, if we have the eigenvalues λ_ℓ and the eigenfunctions $\mathbf{f}_\ell$ of the operators appear in the above equations, (actually, these operators share same eigenvalues and eigenfunctions (Gao and Caines, 2021)) we can transform the equations (16), (17) into the following set of equations to that in $\mathbb{R}^n$:

$$\begin{aligned} \dot{\bar{x}}_\ell(t) &= (A - BR_1(t)^{-1}B^T\Pi(t) + D_0)\bar{x}_\ell(t) \\ &\quad - BR_1(t)^{-1}B^T s_\ell(t) + B\xi_\ell(t) + \lambda_\ell D\bar{x}_\ell(t), \\ \bar{x}_\ell(0) &= \langle \mathbf{m}, \mathbf{f}_\ell \rangle. \end{aligned} \quad (22)$$

$$\begin{aligned} \dot{s}_\ell(t) &= - (A - BR_1(t)^{-1}B^T\Pi(t))^T s_\ell(t) \\ &\quad + \left(\Pi(t)B(R_1(t) + R_2(t))^{-1}R_2(t)R_1(t)^{-1}B^T\Pi(t) \right. \\ &\quad \left. + QH_0 - \Pi(t)D \right) \bar{x}_\ell(t) + \lambda_\ell (QH - \Pi(t)D) \bar{x}_\ell(t) \\ &\quad - \Pi(t)B\xi_\ell(t) + Q\eta\langle \mathbf{1}, \mathbf{f}_\ell \rangle. \\ s_\ell(T) &= -Q_T H_0 \bar{x}_\ell(T) - \lambda_\ell Q_T H \bar{x}_\ell(T) - Q_T \eta \langle \mathbf{1}, \mathbf{f}_\ell \rangle. \end{aligned} \quad (23)$$

After writing their decomposed equations, we have the following proposition.

Proposition 6. If the global forward-backward ODE (17) (16) has a unique classical solution, then it satisfies (22) and (23).

4. VALUE FUNCTION IN INFINITE HORIZON EXPONENTIALLY DISCOUNTED COST CASE

In order to obtain a formula for the value function illustrating the comparative effect of $R_1(t)$ and $R_2(t)$, we employ a similar setting and approach in (Tchuendom et al., 2022).

Assume all the variable in the model are scalars. Let $A = 0, D_0 = 0, D = 0, H_0 = 0, H = 1, \eta = 0, Q = 1$, and corresponding uppercase letters are changed into lowercase letters. Specifically, the model reduces to

$$dx_\alpha(t) = bu_\alpha(t) + \sigma dw_\alpha(t), \quad x_\alpha(0) = x_{\alpha,0} \sim \mathcal{N}(m_\alpha, \nu^2) \quad (24)$$

$$\begin{aligned} J_\alpha(x_{\alpha,0}, u_\alpha(\cdot); z_\alpha(\cdot), \bar{u}_\alpha(\cdot)) &= \mathbb{E} \left[\int_0^\infty e^{-\rho t} \left((x_\alpha(t) - z_\alpha(t))^2 \right. \right. \\ &\quad \left. \left. + r_1(u_\alpha(t))^2 + r_2(u_\alpha(t) - \bar{u}_\alpha(t))^2 \right) dt \middle| x_\alpha(0) = x_{\alpha,0} \right]. \end{aligned} \quad (25)$$

This special case enables an analytical characterization of the way the relative magnitudes of r_1 and r_2 shape the equilibrium behavior through their influence on the value function.

Because we are considering infinite horizon performance, the coefficient of the quadratic term in the following Ansatz for the value function is a constant,

$$V^\alpha(t, x_\alpha) = \Gamma x_\alpha^2 + 2s_\alpha(t)x_\alpha + q_\alpha(t). \quad (26)$$

The value of Γ is determined by an algebraic Riccati equation,

$$\frac{1}{r_1 + r_2} b^2 \Gamma^2 + \rho \Gamma = 1, \quad (27)$$

and its value is taken to be the positive root as in,

$$\Gamma = -\frac{\rho(r_1 + r_2)}{2b^2} + \sqrt{\frac{\rho^2(r_1 + r_2)^2}{4b^4} + \frac{r_1 + r_2}{b^2}}. \quad (28)$$

By applying the Verification Theorem of stochastic optimal control, we get the following proposition stating how to obtain optimal control.

Proposition 7. The optimal control law for the problem determined by (24) and (25) is provided by the solution to the ODE system,

$$\begin{aligned} \dot{\bar{x}}_\alpha(t) &= -\frac{1}{r_1} b^2 \Gamma \bar{x}_\alpha(t) - \frac{1}{r_1} b^2 s_\alpha(t), \\ \bar{x}_\alpha(0) &= m_\alpha, \quad t \in [0, \infty), \end{aligned} \quad (29)$$

$$\begin{aligned} \dot{s}_\alpha(t) &= \frac{1}{r_1} b^2 \Gamma s_\alpha(t) + b^2 \Gamma^2 \frac{r_2}{r_1(r_1 + r_2)} \bar{x}_\alpha(t) + z_\alpha(t) \\ &\quad + \rho s_\alpha(t), \quad t \in [0, \infty). \end{aligned} \quad (30)$$

Then, the optimal control is given by

$$\begin{aligned} u_\alpha^o(t) &= -\frac{1}{r_1 + r_2} b \Gamma x_\alpha(t) - \frac{r_2}{r_1(r_1 + r_2)} b \Gamma \bar{x}_\alpha(t) \\ &\quad - \frac{1}{r_1} b s_\alpha(t), \quad t \in [0, \infty). \end{aligned} \quad (31)$$

These formulas are a particularly degenerate form of those in Proposition 6. Furthermore, we obtain an ODE for $q_\alpha(t)$ by an application of the verification theorem to the problem, it is stated in the following proposition.

Proposition 8. The process $q_\alpha(t)$ satisfies the following ODE,

$$\begin{aligned} \dot{q}_\alpha(t) &= -\sigma^2 \Gamma + \frac{1}{r_1} b^2 s_\alpha(t) - \frac{r_2}{r_1(r_1 + r_2)} b^2 \Gamma^2 \bar{x}_\alpha(t)^2 \\ &\quad + \rho q_\alpha(t) - z_\alpha(t)^2, \quad t \in [0, \infty). \end{aligned} \quad (32)$$

The expectation operator in the cost functional integrates out the randomness of the initial state $x_\alpha(0) \sim \mathcal{N}(m_\alpha, \nu^2)$. Therefore, the overall expected cost at $t = 0$ resolves to a deterministic scalar, given by:

$$V^\alpha(0) = \Gamma(\nu^2 + m_\alpha^2) + 2s_\alpha(0)m_\alpha + q_\alpha(0). \quad (33)$$

A technique for solving the system consists of (29), (30), and (32) is to assume that we are in the stationary state, where the processes are $\bar{x}_\alpha^\infty, s_\alpha^\infty, q_\alpha^\infty$ and satisfy the following equations,

$$0 = -\frac{1}{r_1} b^2 \Gamma \bar{x}_\alpha^\infty - \frac{1}{r_1} b^2 s_\alpha^\infty, \quad (34)$$

$$0 = \frac{1}{r_1} b^2 \Gamma s_\alpha^\infty + b^2 \Gamma^2 \frac{r_2}{r_1(r_1 + r_2)} \bar{x}_\alpha^\infty + z_\alpha^\infty + \rho s_\alpha^\infty, \quad (35)$$

$$\begin{aligned} 0 &= -\sigma^2 \Gamma + \frac{1}{r_1} b^2 s_\alpha^\infty - \frac{r_2}{r_1(r_1 + r_2)} b^2 \Gamma^2 \bar{x}_\alpha^\infty(t)^2 + \rho q_\alpha^\infty \\ &\quad - (z_\alpha^\infty)^2. \end{aligned} \quad (36)$$

From the first two equations, we obtain

$$\left[\left(\frac{b^2 \Gamma^2}{r_1 + r_2} + \rho \Gamma \right) \mathbb{I} - \mathbb{M} \right] \bar{x}_\alpha^\infty = 0, \quad (37)$$

and by the relation in the algebraic Riccati equation, we have $x_\alpha^\infty = 0$. And further we have $s_\alpha^\infty = 0$ $q_\alpha^\infty = \frac{\sigma^2 \Gamma}{\rho}$.

If we write the above variables into their spectral decomposition, we obtain the following proposition giving the coefficients of each component.

Proposition 9. $\bar{x}_\alpha(t)$ and $s_\alpha(t)$ possess the following spectral decomposition,

$$\bar{x}_\alpha(t) = \sum_{\ell=1}^{\infty} \mathbf{f}_\ell(\alpha) \bar{x}_\ell(t), \quad s_\alpha(t) = \sum_{\ell=1}^{\infty} \bar{\lambda}_\ell \mathbf{f}_\ell(\alpha) \bar{x}_\ell(t), \quad (38)$$

where for $\ell \in \{1, 2, \dots\}$, $\bar{\lambda}_\ell$ and $\bar{x}_\ell(t)$ are given by

$$\bar{\lambda}_\ell = \frac{\frac{\rho^2(r_1+r_2)r_2}{2r_1b^2} - \frac{\rho(r_1+r_2)r_2}{r_1b^2} \tilde{\theta}(0) + \frac{r_2}{r_1} + \lambda_\ell}{\frac{\rho r_2}{2r_1} - \theta(\lambda_\ell) - \frac{r_1+r_2}{r_1} \tilde{\theta}(0)}, \quad (39)$$

$$\bar{x}_\ell(t) = \langle \mathbf{m}, \mathbf{f}_\ell \rangle \exp \left[\left(\frac{\rho}{2} - \theta(\lambda_\ell) \right) t \right]. \quad (40)$$

And the functions $\theta, \tilde{\theta}$ are given by

$$\theta(\tau) = \sqrt{\frac{\rho^2}{4} + \frac{b^2}{r_1}(1-\tau)}, \quad (41)$$

$$\tilde{\theta}(\tau) = \sqrt{\frac{\rho^2}{4} + \frac{b^2}{r_1+r_2}(1-\tau)}. \quad (42)$$

Proof. In the infinite horizon exponentially discounted cost function case, for each index $\ell \in \{1, 2, \dots\}$, we have similar ODEs for $\bar{x}_\ell(t)$ and $s_\ell(t)$ to those in Proposition 6,

$$\begin{aligned} \frac{d\bar{x}_\ell(t)}{dt} &= -\frac{1}{r_1} b^2 \Gamma \bar{x}_\ell(t) - \frac{1}{r_1} b^2 s_\ell(t), \quad t \in [0, \infty) \\ \bar{x}_\ell(0) &= \langle \mathbf{m}, \mathbf{f}_\ell \rangle, \end{aligned} \quad (43)$$

$$\begin{aligned} \frac{ds_\ell(t)}{dt} &= \frac{1}{r_1} b^2 \Gamma s_\ell(t) + b^2 \Gamma^2 \frac{r_2}{r_1(r_1+r_2)} \bar{x}_\ell(t) \\ &\quad + \lambda_\ell \bar{x}_\ell(t) + \rho s_\ell(t), \quad t \in [0, \infty), \quad s_\ell^\infty = 0. \end{aligned} \quad (44)$$

By the variation of constants formula, and the relation in the algebraic Riccati equation (27) we have,

$$\begin{aligned} s_\ell(t) &= - \int_t^\infty \exp \left(\left(\frac{b^2 \Gamma}{r_1(r_1+r_2)} + \frac{1}{\Gamma} \right) (t-s) \right) \\ &\quad \left(b^2 \Gamma^2 \frac{r_2}{r_1(r_1+r_2)} + \lambda_\ell \right) \bar{x}_\ell(s) ds, \quad t \in [0, \infty). \end{aligned} \quad (45)$$

Take derivative with respect to t on both sides of (19), substitute (45) into it, and using the relation in (27) we obtain,

$$\frac{d^2 \bar{x}_\ell(t)}{dt^2} - \rho \frac{d\bar{x}_\ell(t)}{dt} + \frac{b^2}{r_1} (\lambda_\ell - 1) \bar{x}_\ell(t) = 0. \quad (46)$$

The characteristic equation of the ODE is given by,

$$\mu_\ell^2 - \rho \mu_\ell + \frac{b^2}{r_1} (\mu_\ell - 1) = 0. \quad (47)$$

Take the negative solution,

$$\mu_\ell = \frac{\rho}{2} - \sqrt{\frac{\rho^2}{4} + \frac{b^2}{r_1}(1-\lambda_{\ell l})} = \frac{\rho}{2} - \theta(\lambda_\ell) < 0. \quad (48)$$

Thus we have the trajectories for $\bar{x}_\ell(t)$, $\ell \in \{1, 2, \dots\}$,

$$\bar{x}_\ell(t) = \langle \mathbf{m}, \mathbf{f}_\ell \rangle \exp \left(\frac{\rho}{2} - \theta(\lambda_\ell) \right), \quad t \in [0, \infty). \quad (49)$$

Substituting the $\bar{x}_\ell(t)$ trajectories (49) into (20), and using the relation in (27), we obtain

$$\begin{aligned} s_\ell(t) &= -\langle \mathbf{m}, \mathbf{f}_\ell \rangle \exp \left(\left(\frac{1}{r_1} b^2 \Gamma + \rho \right) t \right) \\ &\quad \left(b^2 \Gamma^2 \frac{r_2}{r_1(r_1+r_2)} + \lambda_\ell \right) \int_t^\infty \exp \left(\left(\mu_\ell - \frac{b^2}{r_1} \Gamma - \rho \right) s \right) ds \\ &= -\langle \mathbf{m}, \mathbf{f}_\ell \rangle \left(b^2 \Gamma^2 \frac{r_2}{r_1(r_1+r_2)} + \lambda_\ell \right) \\ &\quad \left(\mu_\ell - b^2 \Gamma \frac{r_2}{r_1(r_1+r_2)} - \frac{1}{\Gamma} \right)^{-1} \exp(\mu_\ell t) \\ &= -\langle \mathbf{m}, \mathbf{f}_\ell \rangle \frac{\frac{\rho^2(r_1+r_2)r_2}{2r_1b^2} - \frac{\rho(r_1+r_2)r_2}{r_1b^2} \tilde{\theta}(0) + \frac{r_2}{r_1} + \lambda_\ell}{\frac{\rho r_2}{2r_1} - \theta(\lambda_\ell) - \frac{r_1+r_2}{r_1} \tilde{\theta}(0)} \\ &\quad \exp(\mu_\ell t), \quad t \in [0, \infty), \end{aligned} \quad (50)$$

which completes the derivation. $\square$

Remark 10. Since we are working in the space $L^2[0, 1]$, the appearing countably infinite sum converge at $\alpha \in [0, 1]$, a.e.

Remark 11. If we have the relations, $r_1 + r_2 = 1$, $\delta := r_2/r_1$, then the coefficients of $s_\ell(t)$ will have a cleaner form,

$$\bar{\lambda}_\ell(\delta) := -\frac{\delta b^2 \Gamma^2 + \lambda_\ell}{\frac{\rho}{2} + (1+\delta)b^2 \Gamma + \sqrt{\frac{\rho^2}{4} + b^2(1+\delta)(1-\lambda_\ell)}}. \quad (51)$$

The following proposition gives the explicit formula for the value function.

Proposition 12. The expression for the value function at the initial time 0 is given by

$$\begin{aligned} V^\alpha(0) &= \Gamma \nu^2 + \Gamma(m_\alpha)^2 + 2m_\alpha \sum_{\ell=1}^{\infty} \bar{\lambda}_\ell \langle \mathbf{m}, \mathbf{f}_\ell \rangle \mathbf{f}_\ell(\alpha) + \frac{\sigma^2 \Gamma}{\rho} \\ &\quad + \frac{1}{\rho} \left(\sum_{\ell=1}^{\infty} \lambda_\ell \langle \mathbf{m}, \mathbf{f}_\ell \rangle \mathbf{f}_\ell(\alpha) \right)^2 - \frac{b^2}{r_1 \rho} \left(\sum_{\ell=1}^{\infty} \bar{\lambda}_\ell \langle \mathbf{m}, \mathbf{f}_\ell \rangle \mathbf{f}_\ell(\alpha) \right)^2 \\ &\quad + \frac{r_2}{\rho r_1(r_1+r_2)} b^2 \Gamma^2 \left(\sum_{\ell=1}^{\infty} \langle \mathbf{m}, \mathbf{f}_\ell \rangle \mathbf{f}_\ell(\alpha) \right)^2 \\ &\quad - \sum_{k=1}^{\infty} \sum_{\ell=1}^{\infty} \left(\frac{\rho}{2} - \theta(\lambda_k) \right) (\theta(\lambda_k) + \theta(\lambda_\ell))^{-1} \langle \mathbf{m}, \mathbf{f}_k \rangle \langle \mathbf{m}, \mathbf{f}_\ell \rangle \\ &\quad \mathbf{f}_k(\alpha) \mathbf{f}_\ell(\alpha) \left[\frac{2}{\rho} \lambda_k \lambda_\ell - \frac{2b^2}{\rho r_1} \bar{\lambda}_k \bar{\lambda}_\ell + \frac{2r_2}{\rho r_1(r_1+r_2)} b^2 \Gamma^2 \right]. \end{aligned} \quad (52)$$

The expression (52) can be further simplified as follows.

Proposition 13.

$$\begin{aligned} V^\alpha(0) &= \Gamma \nu^2 + \Gamma(m_\alpha)^2 + 2m_\alpha \sum_{\ell=1}^{\infty} \bar{\lambda}_\ell \langle \mathbf{m}, \mathbf{f}_\ell \rangle \mathbf{f}_\ell(\alpha) + \frac{\sigma^2 \Gamma}{\rho} \\ &\quad - \sum_{k=1}^{\infty} \sum_{\ell=1}^{\infty} \left(\frac{\rho}{\theta(\lambda_k) + \theta(\lambda_\ell)} - 2 \right) \langle \mathbf{m}, \mathbf{f}_k \rangle \langle \mathbf{m}, \mathbf{f}_\ell \rangle \mathbf{f}_k(\alpha) \mathbf{f}_\ell(\alpha) \\ &\quad \left[\frac{1}{\rho} \lambda_k \lambda_\ell - \frac{b^2}{\rho r_1} \bar{\lambda}_k \bar{\lambda}_\ell + \frac{r_2}{\rho r_1(r_1+r_2)} b^2 \Gamma^2 \right]. \end{aligned} \quad (53)$$

To proceed with the analysis, we assume that the initial distributions of the state is uniform in α .

Assumption 14. Assume that $m_\alpha = m, \forall \alpha \in [0, 1]$.

By the above assumption, we have the following simpler formula for the value function.

Proposition 15. Under the assumptions 3 and 14, the following expression holds for the value function,

$$V^\alpha(0) = \Gamma \left(\nu^2 + m^2 + \frac{\sigma^2}{\rho} \right) + 2m^2 \sum_{\ell=1}^{\infty} \bar{\lambda}_\ell \langle \mathbf{1}, \mathbf{f}_\ell \rangle \mathbf{f}_\ell(\alpha) - m^2 \sum_{k=1}^{\infty} \tilde{\lambda}_k^\alpha \langle \mathbf{1}, \mathbf{f}_k \rangle \mathbf{f}_k(\alpha), \quad (54)$$

where $\tilde{\lambda}_k^\alpha$ are given by

$$\tilde{\lambda}_k^\alpha := \sum_{\ell=1}^{\infty} \langle \mathbf{1}, \mathbf{f}_\ell \rangle \mathbf{f}_\ell(\alpha) \left(\frac{\rho}{\theta(\lambda_k) + \theta(\lambda_\ell)} - 2 \right) \left[\frac{1}{\rho} \lambda_k \lambda_\ell - \frac{b^2}{\rho r_1} \bar{\lambda}_k \bar{\lambda}_\ell + \frac{r_2}{\rho r_1 (r_1 + r_2)} b^2 \Gamma^2 \right]. \quad (55)$$

Assumption 16. Assume that $\langle \mathbf{1}, \mathbf{f}_1 \rangle \neq 0$ and $\langle \mathbf{1}, \mathbf{f}_\ell \rangle = 0, \forall \ell \in \{2, 3, \dots\}$.

By reducing the value function to a minimal number of terms, this assumption simplifies the analysis of the effect of δ .

5. NUMERICAL SIMULATION OF TRADE-OFF EFFECTS BETWEEN CONFORMISM AND INDIVIDUALISM

Suppose that we restrict $r_1 + r_2 = 1$, and $\delta := r_2/r_1$. Let the Assumption 16 be in force. Increasing r_1 penalizes individual deviations, while increasing r_2 penalizes deviations from the control mean field, so $\delta := r_2/r_1$ measures the conformism-individualism balance.

We consider scenarios for the following two graphons: Scenario I: Consider the Erdős-Renyi (ER) graph. The only eigenfunctions associated with non-zero eigenvalues is $\mathbf{f}_1(\alpha) = 1$. Although it does not form a basis of $L^2[0, 1]$, it will not invalidate the analysis in this work as $\mathbf{m} \in \text{span}\{\mathbf{f}_1\}$.

Scenario II: Suppose $\mathbf{g}_1$ has the following form,

$$\mathbf{g}_1(\alpha) = \sqrt{0.995} + 0.1 \sin(2\pi\alpha). \quad (56)$$

The two scenarios share the same parameters as follows:

$$b = 0.2, \sigma = 1, \rho = 1, \lambda_1 = 0.1, \nu = 1, m = 1, \Gamma = -\frac{\rho}{2b^2} + \sqrt{\frac{\rho^2}{4b^4} + \frac{1}{b^2}} \approx 0.8114. \quad (57)$$

For the first scenario, numerical simulation shows that there exists an optimal conformism-individualism ratio $\delta_\star$ that minimizes the value function uniformly in α . In contrast, in the second scenario, the optimal ratio $\delta_\star$ varies with the network location α , revealing a non-uniform trade-off that depends on the local network structure.

6. CONCLUSIONS

This work analyzes the existence of the Nash equilibria for LQG-GMFG systems with control mean field costs. These

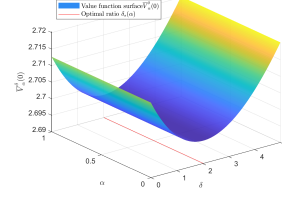


Fig. 1. Scenario I Value function $V^{\alpha, \delta}(0)$

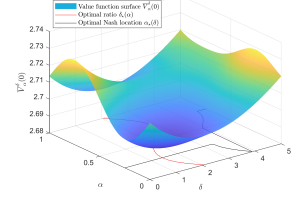


Fig. 2. Scenario II Value function $V^{\alpha, \delta}(0)$

results provide methods and insights for extending the analysis to problems with hard constraints parameterized by the control mean field.

REFERENCES

- Bonnans, J.F., Gianatti, J., and Pfeiffer, L. (2023). A Lagrangian approach for aggregative mean field games of controls with mixed and final constraints. *SIAM Journal on Control and Optimization*, 61(1), 105–134.
- Caines, P.E. and Huang, M. (2021). Graphon mean field games and their equations. *SIAM Journal on Control and Optimization*, 59(6), 4373–4399.
- Cardaliaguet, P. and Lehalle, C.A. (2018). Mean field game of controls and an application to trade crowding. *Mathematics and Financial Economics*, 12(3), 335–363.
- Chen, X. and Wang, Z. (2014). Differential variational inequality approach to dynamic games with shared constraints. *Mathematical Programming*, 146(1), 379–408.
- Conway, J.B. (2019). *A Course in Functional Analysis*, volume 96. Springer.
- Facchinei, F. and Kanzow, C. (2010). Generalized Nash equilibrium problems. *Annals of Operations Research*, 175(1), 177–211.
- Gao, S. and Caines, P.E. (2021). Subspace decomposition for graphon LQR: Applications to VLSNs of harmonic oscillators. *IEEE Transactions on Control of Network Systems*, 8(2), 576–586.
- Gao, S., Caines, P.E., and Huang, M. (2023). LQG graphon mean field games: Analysis via graphon-invariant subspaces. *IEEE Transactions on Automatic Control*, 68(12), 7482–7497.
- Hintermüller, M., Surowiec, T.M., and Theiß, M. (2024). On a differential generalized Nash equilibrium problem with mean field interaction. *SIAM Journal on Optimization*, 34(3), 2821–2855.
- Huang, M., Malhamé, R.P., and Caines, P.E. (2006). Large population stochastic dynamic games: closed-loop McKean-Vlasov systems and the Nash certainty equivalence principle. *Communications in Information and Systems*, 6(3), 221–252.
- Lovász, L. (2012). *Large Networks and Graph Limits*, volume 60. American Mathematical Soc.
- Tchuendom, R.F., Gao, S., and Caines, P.E. (2022). Stationary cost nodes in infinite horizon LQG-GMFGs. *IFAC-PapersOnLine*, 55(30), 284–289.
- Yong, J. and Zhou, X.Y. (1999). *Stochastic Controls: Hamiltonian Systems and HJB Equations*, volume 43. Springer Science & Business Media.